INFO123 Project: Build a Business with AI

Overview

Generative AI tools, such as Claude, Gemini, ChatGPT and Mistral are becoming essential business tools which, if used wisely, can augment human cognitive abilities and productivity. For this project, you will use generative AI to build a breakfast products business from scratch.

You will use generative AI to do the following:

- Create a breakfast product
- Analyse and visualize market data
- Create a company
- Develop a marketing strategy, including product packaging and launch advertisement
- Define KPIs for the core business processes
- Model strategic information systems and core data
- Develop a customer support chatbot
- Develop the business' web page using a *vibe-coding* agent

You will use this information to create a finalized business plan and a video presentation (2 to 3 minutes max). In this presentation, you will pitch your new business to investors and you will discuss your experience in using AI for creating your business.

Deliverables to be submitted to LEARN

- A comprehensive, polished business plan document (a collection of artefacts from the business creation tasks, packaged into a single PDF document) - see task 13
- A short advertisement jingle (15-30 sec., via an audio file .mp3) - see task 8
- A customer support chatbot (a working URL) - see task 11
- A demo of a website (a working URL) - see task 12
- A video presentation of business pitch and lessons learned about working with AI for this assignment (**via a video file .mp4 or .mov only** – no links) - see task 14
- A log of all chat session URLs (**via the submission form on Learn**)

Due Date & Submission

16 October 2026 5pm

Instructions

The pages that follow contain a step-by-step guide for creating the content of the business plan for your breakfast products business. You are expected to be creative, **go beyond these instructions and examples** by applying the techniques covered in the lectures and in the labs, and push the boundaries of AI capabilities.

Essential Workflow Practices

1. Use multiple chat sessions: To avoid “context rot” (when the AI loses track of earlier context in long conversations), create one chat session per task or group of related tasks. For example:

- One chat for market data analysis (Task 2)
- One chat for product development (Tasks 3-4)
- One chat for marketing strategy (Tasks 7-8)
- And so on...

2. Keep track of your work in a project folder on your computer: Set up a dedicated folder on your computer to log and store all content produced by the AI as you progress through the tasks. In this folder, organise:

- Content generated by AI (paste it in text files or Word documents)
- Data analyses and figures
- Images (logos, product packaging, social media ads)
- Audio files (jingles)
- Image files of process diagrams and data models
- Personal notes and reflections: keep track of key decisions you made about your business

You will need these materials to create your final business plan and video presentation.

3. Keep a log of ALL links to your chats: Use your AI tool's share function to generate permanent URLs for each chat session. **You need to submit all chat links via a form on Learn, and flag 2–3 sessions that show your best work with AI.** When choosing which sessions to flag, pick ones that show how you steered the AI and how you checked its claims against a real source. **DOUBLE-CHECK THAT THE URLs ARE ACCESSIBLE**.

4. Save and export everything: Don't rely solely on chat history. Copy or download all outputs locally:

- Export images and diagrams as files
- Copy and paste text into documents
- Save audio files

Chat sessions can be deleted or lost, so maintain your own copies.

Your Tasks

Task 1: Create First Product Ideas

Begin by working with the AI to generate ideas for your first breakfast product, tailored to your preferences. The breakfast products category typically includes 4 sub-categories: breakfast cereals, nutritional cereal bars, spreads, and liquid breakfasts.

Prompt the AI to ask you interactive questions about the kind of breakfast product you want to create to generate product ideas.

Task 2: Analyse current market trends

You obtained some data on past breakfast product trends in New Zealand. Use the AI to identify potential opportunities for launching a new breakfast product.

Download the Excel dataset ***nz_breakfast_sales_2022_2025.xlsx*** from Learn.

1. Upload the dataset to a chat and ask the AI to identify, report, and explain data quality issues.

2. Work with the AI to fix any data quality issues it has found.

3. Ask the AI to analyse and visualise key aspects of the data to inform your market research.

Store the results of each step in your project log and folder, including any figures you have produced with the AI.

Task 3: Finalise your product

Finalise your first product and determine the flavour and recipe.

1. Prompt the AI to combine your breakfast product preferences and the results of the data analysis to determine what you should create for your first breakfast product

2. Brainstorm and identify a name for your first breakfast product.

3. Determine the possible flavour(s) of the breakfast product.

4. Next, create a recipe to produce the breakfast product.

5. The recipe will likely be designed to be made at home; ask the AI to provide an industrial version of this recipe that can be produced at scale and at low cost. You can ask the AI for additional constraints or business requirements that fit your business strategy.

6. Generate a brief product description.

    - Example prompt: *Based on the discussion so far, create a brief product description.*

Record the results of each step in your project folder.

Task 4: Critique and refine your product idea

It's time to get some feedback on your product idea.

1. Create a document that summarises the breakfast product information you produced in Task 3. With your prompt, upload the traces you have collected in Task 3.

    - Example prompt: *Create a document that summarizes my breakfast product, including the recipe, name, and motivation behind the breakfast product. This information will be used to determine the commercial viability of my product.*

2. Upload the summary document into a new chat and ask the AI to assess the various ideas.

    - Example prompt: *Attached is a new breakfast product idea. Analyse the commercial viability across the following dimensions: product differentiation and uniqueness, consumer appeal, brand and marketing potential, and manufacturing feasibility. For each, give the product a score out of 5, explaining your reasoning, and then provide an overall score.*

Record the results of each step in your project folder.

Task 5: Gather intelligence on market competitors

Research the competitive landscape to understand how your product compares to existing offerings in the market.

1. Search online for the nearest competitors.

- Example prompt: *Search online for the competitive landscape, and what are the nearest products to mine? How much do they charge? What is my differentiation?*

2. After you finish the search, be sure to unselect the web search feature to avoid lengthy searches and using up your messaging limits.

3. Ask the AI to analyse your product's differentiation compared to competitors.

4. Identify 2-3 key competitors and record their:

    - Product names and descriptions
    - Pricing
    - Target customers
    - Key selling points
    - Your competitive advantages

Record the results of each step in your project folder.

Task 6: Create a company name and logo

1. It's time to pick a company name. To receive suggestions from the AI, provide some information about your product and what your company should be about. Ask the AI to generate name options and work with the AI to find a name that best fits your vision.

2. Now generate a logo for your company.

    - **Tip 1:** In the AI, click on the "Tools" button and then "Create Image". You can also use one of the free tools we list on p., but your ability to iterate and refine the logo might be more limited because of lower usage limits.

    - **Tip 2:** *Use fast models for prototyping images and thinking models for final output:* A kind reminder to pay attention to your usage limit; generating images may result in faster model usage. If you generate too many images, you may exceed the usage limit and be unable to generate any more. Use cheap and faster generation with smaller (non-thinking) models to develop and iterate your image prompt. Once you have locked in your image design, then use the *thinking* or *pro* mode to generate a higher quality output. The thinking mode will use up your credits faster.

    - **Note:** Image generation often contains spelling mistakes, as the model generates pixels rather than words written on an image. Models are improving quickly, but this remains a limitation of existing image generation technology.

Save the results of each step in your project log and folder.

Task 7: Target customer and packaging

After developing the initial product, identify a target customer segment and develop a packaging prototype.

1. Begin by identifying a customer segment that is most likely to purchase the breakfast product. A customer segment is a specific group of potential buyers who share similar characteristics, needs, or behaviours; they are the most likely to purchase the breakfast product.

2. Based on this information about the customer segment, generate an image of a *product package design*. This should be an image that realistically depicts the product's physical appearance, showcasing how it would look to a consumer in a real-world retail environment.

3. Get feedback: ask the AI to provide *synthetic* customer feedback on your ideas. In a new chat, upload the product package image, provide the information you want feedback on, and ask the AI to take on a persona that represents your target customer segment.

    - **Tip:** A common strategy for improving the outputs is to specify a persona for the AI. Instead of just asking the AI “You are a customer...” describe a person with specific goals and hurdles, such as “You are a time-crunched parent looking for low-sugar options” or “You are a fitness enthusiast focused on sustainable packaging.”

4. Incorporate the feedback into your product package design and regenerate an image for it.

Save the results of each step in your project folder.

Task 8: Online marketing strategy

With the information you have gathered so far, create a marketing strategy that includes ideas for social media and podcast advertisement.

1. Provide the AI with the content and images you created in Task 4, 5 and 7. Ask the AI to create an initial text-based product description for promoting the product. Tell it to limit the description to 250 words.

2. Ask the AI to criticize its work. Based on its suggestions, improve the draft product description.

- **Tip:** You can provide more information about the type of tone and other details that you are looking for by, for example, adding the following to your prompt: *Use a more formal tone* **or** *use a more playful tone.*

3. Design a potential social media advertisement for this product.

    - Start by ask the AI to identify the appropriate channels: *What social media channels are best suited to reach the product's target audience?*

    - Ask the AI to describe an advertisement you could create to promote this product.

4. Generate the image of a social media ad based on this advertisement design and your promotional product description. If the image generation tool allows you to use images as input (e.g. Gemini or ChatGPT), also add your company logo and product packaging image to your prompt.

5. Create a 15- to 30-second jingle/song to advertise your product on podcasts.

    - Example prompt: *Create a 15-second jingle to go with the advertising campaign.*

    - **Tip:** In your prompt, specify the *vibe* and *genre* of the music: energetic, relaxed, slow-tempo, fast-tempo, pop, jazz, rock, female or male vocalist, etc.

    - **Note 1:** Gemini can generate lyrics and music (in “Tools”). Claude, ChatGPT and Mistral will just give you the lyrics; to generate an actual song you will have to use other tools, such as ElevenLabs, Suno, or Udio.

    - **Note 2:** The AI will likely NOT follow your time limit instructions - that is a known limitation of AI music generators. That is okay: aim for 15-30 seconds and work with the AI to keep the jingle length to under a minute maximum.

6. Ask the AI to identify New Zealand podcasts that would be good channels to advertise your product to your target customers.

7. Combine everything in a marketing strategy document.

    - Example prompt: *Based on the discussion so far, generate a marketing strategy document.*

Save the outputs of each step in your project folder.

Task 9: Operations Strategy and Core Business Processes

Now that you've developed your product and marketing strategy, it's time to think about how your business will actually operate. You'll identify a manufacturer and map out your most critical business process.

1. Identify a company to make your breakfast products in New Zealand. Be sure to enable the web search option so the tool can find real companies. Provide the AI with enough context from your previous tasks outputs so that it provides you with relevant findings.

    - Example prompt: *Find a company that can manufacture my breakfast product in New Zealand.*

2. Record the name and contact details of at least one potential manufacturer.

3. Work with the AI to identify the key business processes for your breakfast products company.

    - Example prompt: *What are the core business processes I need to operate my breakfast products company? Consider processes like order fulfilment, inventory management, customer service, and product distribution.*

4. Select the one process that is most critical to the success of your business and ask the AI to create a process flow diagram using Mermaid notation.

    - Example prompt: *Create a process flow diagram in Mermaid format showing the [name of process] for my breakfast products business. This is the most critical process for my business success.*

    - **Tip:** Ask *explicitly* the AI to generate a Mermaid diagram. You can copy the Mermaid code and paste it into a Mermaid live editor (https://mermaid.live/) to view and export the diagram as an image. If Mermaid.live gives you an error when rendering your diagram based on the AI generated code, copy-paste the error code in the chat and ask the AI to fix it.

5. Review the process diagram and ask the AI to identify potential bottlenecks or inefficiencies.

    - Example prompt: *Looking at this process flow, what are the potential bottlenecks or risks? What could go wrong?*

Record the results of each step in your project log and folder, including:

- Manufacturer name and contact details
- Core business processes
- Process flow diagram of one critical business process (as image)
- Analysis of potential bottlenecks

Task 10: Model Your Information Needs

Every business needs to track and manage information to operate effectively. In this task, you'll identify what data your business needs and how different pieces of information relate to each other.

1. Work with the AI to identify what information your business needs to track and store to operate effectively, including key performance indicators (KPIs) you would monitor. Don't forget to provide the AI with enough context from your previous tasks outputs.

    - Example prompt: *What are the key data entities my breakfast products business needs to track? Think about customers, products, orders, inventory, suppliers, etc. Also identify the key performance indicators (KPIs) I should monitor to measure business performance.*

2. Ask the AI to create an Entity Relationship Diagram (ERD) in Mermaid format showing how these data entities relate to each other.

    - Example prompt: *Create a simple Entity Relationship Diagram in Mermaid format showing the key data entities for my business and how they relate to each other. Include entities like Products, Customers, Orders, and Inventory.*

    - **Note:** This is a simplified business-level data model, not a technical database design. Focus on understanding what information your business needs and how different pieces of information connect. You can copy the Mermaid code and paste it into a Mermaid live editor (https://mermaid.live/) to view and export the diagram as an image. If Mermaid.live gives you an error when rendering your diagram based on the AI generated code, copy-paste the error code in the chat and ask the AI to fix it.

3. Ask the AI to explain how this data structure supports your business operations and KPIs.

    - Example prompt: *Explain how this data model would help me run my business. What questions could I answer with this information? What business decisions would this data support? How would I track the KPIs you identified?*

Record the results of each step in your project log and folder, including:

- List of key data entities and KPIs
- Entity Relationship Diagram (as image)
- Explanation of how the data model supports business operations

Task 11: Create a Customer Support Chatbot

Create a customer support chatbot that can answer common questions about your breakfast products business.

1. Use your AI tool's agent/chatbot creation feature to build a customer support chatbot for your breakfast products business.

    - **For Gemini users:** Use the *Gem* feature (in the left-side menu → *Create Gem*) to create a custom chatbot with specific instructions about your product, company values, and how to handle common customer questions.

    - **For Mistral users:** Use the *Agent* feature to create a custom agent with instructions about your business and products.

    - **For ChatGPT users:** Use the *CustomGPT* feature (via https://chatgpt.com/gpts or in the left-side menu → *Explore GPTs* → *+Create* button in the top right)

    - **For other tools:** Check if your AI tool has a custom GPT, agent, or chatbot creation feature.

2. In the system prompt, configure your chatbot with information about:

    - Your company name and values
    - Product details (name, ingredients, benefits)
    - Common customer questions (where to buy, nutritional information)
    - Your target customer segment
    - Tone of voice (friendly, professional, playful - whatever matches your brand)

    - **Tip:** In a separate chat, ask the AI to generate a system prompt for your chatbot. Example prompt for creating chatbot instructions: *Create a customer support chatbot system prompt for my breakfast products company [Company Name]. The chatbot should be able to answer questions about [Product Name], including ingredients, nutritional information, where to purchase, and company values. The tone should be [describe tone - friendly, professional, etc.]. Here are the key details: [paste product information].*

    - **Tip:** You can upload to the *knowledge base* of the chatbot relevant outputs that you generated from the previous tasks. The chatbot will use the knowledge base when answering customer questions.

3. Test your chatbot by asking it several customer questions and refining its instructions based on the responses.

4. Make your chatbot publicly accessible and save the public URL. **Test the URL in a private/incognito browser window to ensure it is publicly accessible (or ask a friend to check it).**

    - **Important:** You must submit a working public URL for your chatbot. Make sure anyone can access it (it's okay if we have to login to Gemini).

Record the chatbot's public URL in your project log. You will need to submit it to Learn. Test that it works!

Task 12: Generate a Website Demo

Create a demo website for your breakfast products business. The website must have a landing page at minimum.

1. Choose your website building approach:

    - **Figma Make** (Recommended): Generous free education plan, good quality results: https://www.figma.com/education/apply
    - **AI tool built-in preview**: Gemini, ChatGPT or Mistral's code preview (temporary URLs)
    - **Alternative vibe-coding tools**: Replit, v0, Lovable, Manus (see AI Tools section)

2. Ask the AI to generate the website based on your branding. Provide it with the relevant outputs from the previous task (e.g., product description and packaging, company name and logo, online marketing strategy). The website should include at minimum a landing page. If you want to go further, you can add a product page and a shopping cart:

    - **Landing page**: Introduces your company and product - REQUIRED
    - **Product page**: Displays product description and images - OPTIONAL
    - **Shopping cart/checkout page**: Shows purchasing flow - OPTIONAL

3. Generate a permanent, publicly accessible URL.

    - For Figma Make: Use the share/publish feature
    - For other tools: Follow their sharing instructions
    - **Important: The URL must work without requiring login (double-check!)**

4. Test the URL in a private/incognito browser window to confirm public accessibility.

Record the website URL in your project log and test that it works! You will need to submit it to Learn along with your business plan.

Task 13: Summarise into a Business Plan (PDF format)

Work with the AI to create a business plan of the breakfast product proposal, market & competitor analysis, marketing strategy, operations & information systems strategy, and analysed data.

Important: ***The business plan should be ready to present "as-is" to real investors*****:** accurate and reliable, fact-checked, as well as professional looking in style and format.

The business plan should between 1500 to 2500 words (this is not a strict word count limit but keep the length within reasonable range). This word count doesn't include images, figures or charts. If your document is too long, work with the AI to shorten it.

1. Summarise the discussion so far:

    - Example prompt: *Based on the documents I provide you, create a concise business plan document that encompasses all my research and content for this business idea including, but not limited to, the breakfast product proposal, marketing strategy, operations and information strategy, and analysed data. The document should also include an executive summary, as it will be used to pitch this company and product to potential funders. The document should contain 1500 to 2500 words maximum.*

2. **Important**: The business plan document must contain the following outputs you have already produced:

    - One or more figures from the market research (task 2)
    - The company logo (task 6)
    - The product packaging image (task 7)
    - The social media advertisement image (task 8)
    - The critical business process diagram (task 9)
    - The ERD diagram (task 10)

3. The document should be submitted to LEARN in a **PDF format**.

Note: most AI tools still struggle to produce PDF documents directly. This step requires you to get involved in the creation and editing of the document; you will likely need to use Microsoft Word and Google Docs to do the final conversion to PDF. We expect you to steer the creation of the business plan, and make sure it has your "personal touch": that it doesn't read as "inauthentic" and "sloppy."

Task 14: Video Presentation

Create a 2-3 minute video presentation in talking head format where you pitch your business and reflect on your experience using AI for this project. You will record yourself presenting using PowerPoint's built-in recording feature.

- **Recording format**: Talking head (you speaking to camera, with slides visible)
- **Duration**: We expect 3 minutes maximum (hard stop at 3m29s; we stop listening if longer and the excellence criteria can't be attained – please don't ask if 3m30s is okay, the answer will be no.)
- **Recording tool**: PowerPoint's built-in "Record Slide Show" feature (or similar screen recording with webcam); use *File → Export As → Format: .mp4* to export the video.

Important: The video presentation has two parts: (part 1) the business pitch (1 minute max) and (part 2) your personal reflection about AI (1 to 2 minutes).

Part 1: Business Pitch (1 minute max – strict limit)

Create 2 slides to pitch your breakfast products business to potential investors. You may use AI to help generate the content for these slides, but you must review and personalise the content before using it. Make the content yours. The title slide for your presentation doesn't count among these 2 slides.

- Example prompt: *Based on my business plan, generate content for two presentation slides I can use to pitch my breakfast products business to potential funders. The slides should highlight the product, market opportunity, and competitive advantage. Please carefully review each slide to ensure accuracy.*

- **Note:** While Gemini, Mistral, and ChatGPT can generate slides and content via the Canvas feature over the web, they do not provide direct conversion to PowerPoint. Mistral creates slides in Markdown format (.md, a text file) and Gemini slides can be opened in Google Slides. You will have to transfer their content to PowerPoint to record yourself. There are other tools, such as Gamma and Manus, that can make high-quality slides (but they are rate-limited). The desktop apps of Claude & ChatGPT also work directly with PowerPoint via a plugin.

2. Transfer the AI-generated content into PowerPoint slides. Review and refine the content to ensure it accurately represents your business.

3. Your 1-minute maximum (strict limit) pitch should cover:

    - Your product and what makes it unique
    - The market opportunity (based on your data analysis)
    - Your competitive advantage

Part 2: Personal Reflection (1-2 minutes)

IMPORTANT: This part must be entirely your own work. Do NOT use AI to write your reflection or slide content for this section. This is the only task in the assignment where AI assistance is strictly not permitted.

Create 2 to 4 polished and well-designed slides for your reflection. Discuss the following two topics in your video:

Reflection Question 1: Your lessons about working with AI
Reflect on your experience working with generative AI throughout this project. What were the most important lessons you learned about prompting, context engineering, session management, fact checking, validation and working effectively with AI tools?
Describe a specific moment when the AI either significantly helped you or caused issues, and explain how you handled it. What would you do differently if you were to complete a similar project again? Be specific and avoid vague generalities.

Reflection Question 2: Critical Evaluation of AI Content
Throughout this project, you used AI to generate business content, analyse data, create marketing materials, model processes and information systems, and build prototypes. Reflect critically on the authenticity, quality, and ethics of AI-generated content. Here are some suggestions for what you can explore in your discussion:
How did you ensure the AI content felt “real”? How did you avoid creating “AI slop” (generic, low-quality AI content)? What are the limitations and risks of relying on AI for creating a business? Does your use of AI in creating a business raise ethical concerns? What is right or wrong about such use? Think of the implications for you personally as a future knowledge worker and what you could do about those. It’s preferable to do a deep critique on specific aspects of your experience than a broad critique on many aspects.

Your reflection should:

- Be honest and personal
- Include **specific examples** from your project
- Demonstrate critical thinking about AI's capabilities and limitations
- Show what you learned beyond just how to use the tools

Recording Your Presentation

1. Open your PowerPoint presentation with all slides (pitch + reflection).

2. Use PowerPoint's "Record Slide Show" feature:

    - Go to the "Slide Show" tab
    - Click "Record Slide Show"
    - Enable your webcam and microphone
    - Record yourself presenting each slide

3. Export your recording as a video file (in .mp4 format):

    - Go to File → Export As → File Format → .mp4
    - Choose video quality (HD recommended)
    - Save the file and upload it to Learn (submit the file, not a link to the file)

4. Alternatively, you may use any screen recording software with webcam overlay to record your presentation, *but you still need to submit a video file to Learn (not a link).*

Presentation tips:

- Practice your presentation and reflection before recording
- Speak clearly and at a moderate pace
- Look at the camera when speaking
- Avoid reading from a script
- Keep to the 2-3 minute time limit
- Ensure good lighting and audio quality
- Be professional but authentic

Submission

Mandatory: Upload your video presentation to Learn as a video file (.mp4 or .mov).

- **Do not submit a shared link from a web service** (OneDrive, YouTube, Dropbox). **You must submit the original video file to Learn**. Links can't break or become inaccessible – this is the only way to ensure that we can mark your video.

AI Tools

Required Setup (Action Required)

Before starting the assignment, complete the following:

Figma Studio Education Account (Free, for Task 12 Website Creation)

- Register at: https://www.figma.com/education/higher-education/
- Use your @uclive.ac.nz email address
- Complete student verification process

We **strongly recommend** subscribing to one of the following AI assistants as your primary tool for this assignment. A subscription provides:

- Higher usage limits for iteration and refinement
- Integrated image generation

- Built-in code/website preview capabilities
- Data analysis features

Option 1: Google Gemini AI Plus (Recommended)

- Cost: **Free for one year** (as of September 1, 2026 – check the link below; this offer could change at any time).
- Regular cost: $8.99 NZD/month
- Features: Multimodal chat, image & music generation, Canvas for slides, code interpreter, Gem creation for chatbots, Stitch for website creation
- Student plan: https://gemini.google/students/
- Subscribe: https://gemini.google/subscriptions/

Option 2: Mistral Le Chat Pro (Good value alternative)

- Cost: $5.99 USD (~$10 NZD)/month (as of September 1, 2026)
- Features: Capable models, image generation, Canvas for slides, Agent creation for chatbots
- Subscribe: https://chat.mistral.ai/upgrade/student
- **Use your @uclive.ac.nz email when signing up**

Option 3: ChatGPT Go

- Cost: $8 USD (~$14 NZD)/month (as of September 1, 2026)
- Subscribe: https://chatgpt.com/pricing
- **Note:** ChatGPT will introduce ads during 2026. We recommend Gemini or Mistral instead.

Option 4: Claude Pro

- Cost: $17 USD (~$30 NZD)/month (as of September 1, 2026)
- Subscribe: https://claude.ai/pricing
- **Note:** Claude is a frontier AI tool with excellent capabilities, but it is expensive in comparison to the options above. Gemini and Mistral have sufficient capabilities to complete the assignment – it's your prompting and workflow that is assessed, not the AI.

Already have a subscription? If you already have Claude Pro, Gemini AI Pro, ChatGPT Plus, or another premium AI subscription, you do not need to switch.

Tools by Task Category

For General Chats

- **Primary**: Gemini, Mistral, or ChatGPT
- **Supplementary**: Duck.ai, Claude, Perplexity, HuggingChat, Microsoft CoPilot

For Data Analysis (Task 2)

- **Primary**: Gemini, Mistral, ChatGPT
- **Supplementary**: Claude.

For Image Generation (Tasks 6, 7, 8)

Gemini, Mistral and ChatGPT have integrated image generation tools.

Standalone free tools (if you need more generations or your primary tool hits usage limits):

- ElevenLabs: https://elevenlabs.io/app/image-video (limited free tier)
- Stability.ai: https://dreamstudio.stability.ai/ (limited free tier)
- Luma DreamMachine: https://dream-machine.lumalabs.ai/ (limited free tier)

Tip: Image generation consumes usage limits quickly. Start early and plan your iterations carefully; avoid generating dozens of variations.

For Audio Generation (Task 8 - Jingle/Song)

Gemini can create both lyrics and music. Mistral, Claude, ChatGPT will generate lyrics only. Other tools that can generate audio include:

- **ElevenLabs**: https://elevenlabs.io/ - Text-to-speech and music generation (limited free tier, has a studio feature that allows you to mix a voice track and a music track)
- **Suno**: https://suno.com/ - Music and song generation (limited free tier)
- **Udio**: https://udio.com/ - Music generation (limited free tier)

Note: Free tiers have daily or monthly limits. Start early to avoid last-minute issues.

For Process Diagrams and ERD Models (Tasks 9, 10)

Gemini, Mistral, ChatGPT and Claude can generate Mermaid code.

Rendering Mermaid diagrams:

- **Mermaid Live Editor**: https://mermaid.live/ (Free, no login required)
- Copy AI-generated Mermaid code and paste into the editor. If there are errors in rendering, copy-paste the error messages back to Gemini, Mistral or ChatGPT, and ask it to fix the errors.
- Export as PNG or SVG for your business plan

For Chatbot Creation (Task 11)

- **Gemini**: Use "Gem" feature (left-side menu -> Create Gem)
- **Mistral**: Use "Agent" feature
- **ChatGPT**: Use "Explore GPTs" feature (left-side menu -> +*Create*)

For Website Generation (Task 12)

Recommended (best quality):

- **Figma Make** (Free education account): https://www.figma.com/make - High-quality website prototypes, generous limits for students: https://www.figma.com/education/higher-education/
- **Google Stitch**: (Free) https://stitch.withgoogle.com/ - High-quality mockups of websites, also free, slightly less user-friendly than Figma

Built into AI tools: You can also ask the Gemini, Mistral, Claude and ChatGPT to generate a static web page via their built-in code preview feature. If you ask the AI to code a web page, it will open a "canvas" or "artefact" window that previews the web page. This preview is usually shareable via a public URL.

Alternative vibe coding tools: The following options are powerful tools that will generate beautiful web pages, but their free tier is limited. They require registration.

- Replit: https://replit.com/ (limited free tier available)
- v0: https://v0.app/ (limited free tier)
- Manus: https://manus.im/ (limited free tier)
- Lovable: https://lovable.dev/ (limited free tier)

Important: You can use any of these tools but ***verify that your final website demo has a permanent, publicly accessible URL for submission***.
If we can't access it, we can't mark it.

For Video Presentation (Task 14)

Recording:

- PowerPoint's built-in "Record Slide Show" feature (Recommended)
- Any screen recording software with webcam overlay

Important notes about free AI tools:

- **Usage limits**: Free tools have strict daily/monthly caps that reset at different times
- **Privacy**: Free tools may use your inputs to train future models - do not include sensitive personal information
- **Data retention**: Free tools may have different data retention policies than student/pro plans
- **Reliability**: Usage may be throttled during peak times

General Tips

1. **Plan ahead**: Don't wait until the night before submission to work on this assignment.
2. **Use free tools strategically**: Save them for when you hit usage limits on your primary tool
3. **Test chatbot and website URLs early**: Ensure they're publicly accessible before final submission
4. **Save your work frequently**: Keep copies of all AI-generated content on disk
5. **Export and download**: Don't rely on chat history alone. Save images and content locally

FAQ

Q: Can I edit the AI outputs or should I let the AI do everything?

A: The goal of the project is to develop your skills in using AI effectively. Therefore, the question you should ask yourself is: what approach will likely yield the best outcome? Letting the AI do all the work by itself is unlikely to be the most optimal approach to get a good result out of it. If you think you need to intervene or customise the content while you work with an AI, that's absolutely fine - and it's likely to make the outcome better. That's true for any task you may do with AI, not just the task of creating and putting together a report.

Q: Sometimes when I want to get a sharing link for my Gemini chat transcript the link is greyed out. What should I do?

A: In this case, paste the full text of the chat into a Word document – include both your prompts and Gemini's answers. Submit the Word file along with your video on Learn, and indicate in the chat form that you submitted a file that contains the transcript of your chat.

Q: I created my chatbot as a gem in Gemini. I can't get it to work in incognito mode.

A: You must be logged in to a Google account for a gem to work. To test it, you should log in or share it with another user who is logged in. You do not have to rebuild your chat bot in another model.

Q: I created my chatbot in ChatGPT. I can't get it to work in incognito mode.

A: You must be logged in to ChatGPT for the GPT agent to work. To test it, you should log in or share it with another user who is logged in. You do not have to rebuild your chatbot in another model.

Q: I asked the AI to produce a PDF but it won't do it.

A: If the AI won't produce a pdf directly, ask it explicitly to make a Google Docs or Word document. Then you can edit and save the document yourself as a PDF.

Q: How should I get the AI to insert the images and Mermaid diagram into my plan?

A: You should ask the AI to make a placeholder where the images and diagram need to go in the document. Then you can paste them into that place yourself.

Q: Do I need to submit the project folder mentioned in "Essential Workflow Practices"?

A: No, the brief only suggests creating a folder on your computer to organise your workflow and keep track of the outputs you create with the AI.